FORENSIC AUDIT EXECUTIVE REPORT

Dataset Evaluated: **clean_Financial_Fraud_Dataset.csv** | Status: Complete

Phase 3 Baseline Score	Phase 4 Forensic Score	P3 Flagged	P4 Flagged	Newly Surfaced
44.13 / 100	19.05 / 100	374	119	44

Forensic Test Breakdown

Test Executed	Calculated Risk Score	Records Flagged
Duplicate Detection	0.00	0
Outlier Detection	2.86	6
Correlation Analysis	54.29	114

Executive Findings & Recommendations

Executive Audit Summary

1. **Executive Risk Assessment**
The system exhibits **High Risk** due to a Phase 3 Baseline Risk Score (Score 1) of 44.13/100 and a Phase 4 Targeted Forensic Score (Score 2) of 19.05/100. While Score 1 indicates moderate baseline risk, Score 2's low value highlights critical vulnerabilities in forensic detection. The 374 Phase 3 flagged records and 119 Phase 4 flagged records (with 75 overlaps) underscore significant systemic risks, particularly in correlation analysis and outlier detection.

2. **Key Anomalies Identified**
- **High Customer Risk Scores**: 374 records flagged in Phase 3 include customers with elevated risk scores (e.g., 88) and frequent transactions (e.g., 28 transactions in 24 hours).
- **Merchant Risk Mismatch**: 119 Phase 4 flagged records involve high-risk merchants (e.g., Fuel category with "High" risk level) paired with low-risk merchant risk labels.
- **Anomaly Detection Gaps**: Outlier Detection scored 2.86/100, with only 6 flagged records, suggesting limited efficacy in identifying irregular patterns. Correlation Analysis, scoring 54.29/100, flagged 114 records, indicating systemic issues in transaction pattern linkage.
- **Newly Surfaced Risks**: 44 records flagged in Phase 4 represent previously undetected threats, requiring immediate scrutiny.

3. **Actionable Recommendations for Audit Team**
- **Prioritize Correlation Analysis**: Focus on resolving the 114 flagged records from Correlation Analysis to address systemic transaction pattern mismatches.
- **Audit High-Risk Customers**: Investigate 374 Phase 3 flagged records, particularly those with high customer risk scores (e.g., 88) and frequent transactions.
- **Enhance Outlier Detection**: Improve the Outlier Detection score (currently 2.86/100) by refining thresholds for unusual transaction frequencies or amounts.
- **Validate Merchant Risk Labels**: Re-evaluate 119 Phase 4 flagged records to ensure alignment between merchant risk levels and transaction behavior.
- **Monitor Newly Surfaced Risks**: Conduct a deep dive into the 44 newly surfaced records to identify emerging fraud vectors or system blind spots.
- **Review Authentication and Device Trust**: Flagged records with low device trust scores (e.g., 67) or high failed login attempts (e.g., 3) require closer examination of authentication vulnerabilities.

This summary emphasizes the urgency of addressing both systemic detection gaps and high-risk individual records to mitigate fraud exposure.